

Unit 8 Quiz Study Guide DRAFT

Exponential Functions | Draft New Format

Name: _____

Period: _____ Due: _____

How to use this guide: This draft is built from the Unit 8 quiz/practice-set skills: classifying linear/quadratic/exponential tables, writing $y = ab^x$, interpreting growth/decay, and using regression-style models. Each skill has a Version A and Version B so students can try a second similar problem after getting help.

Multiple-Choice Practice

1A. Classifying Tables | Version A

For y-values 3, 9, 27, 81, what model type fits best?	Work / thinking
A. linear	_____
B. quadratic	_____
C. exponential	_____
D. constant	

1B. Classifying Tables | Version B

For y-values 4, 9, 16, 25, what model type fits best?	Work / thinking
A. linear	_____
B. quadratic	_____
C. exponential	_____
D. constant	

2A. Growth or Decay | Version A

In $y = 80(1.12)^x$, what is the percent rate?	Work / thinking
A. 12% growth	_____
B. 12% decay	_____
C. 112% growth	_____
D. 88% decay	

2B. Growth or Decay | Version B

In $y = 500(0.76)^x$, what is the percent rate?	Work / thinking
A. 76% growth	_____
B. 24% growth	_____
C. 76% decay	_____
D. 24% decay	

3A. Writing Exponential Models | Version A

A table has $y = 6$ when $x = 0$ and multiplies by 4 each step. What is the model? A. $y = 4(6)^x$ B. $y = 6(4)^x$ C. $y = 10x$ D. $y = 6x + 4$	Work / thinking
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3B. Writing Exponential Models | Version B

A table has $y = 20$ when $x = 0$ and multiplies by 0.5 each step. What is the model? A. $y = 0.5(20)^x$ B. $y = 20(0.5)^x$ C. $y = 20x + 0.5$ D. $y = 10^x$	Work / thinking
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4A. Prediction | Version A

For $y = 250(0.8)^x$, what is y when $x = 3$? A. 128 B. 160 C. 200 D. 512	Work / thinking
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4B. Prediction | Version B

For $y = 100(1.5)^x$, what is y when $x = 2$? A. 150 B. 200 C. 225 D. 250	Work / thinking
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Free-Response Practice

Show clear work. Use the Version B problem as a second check after you have tried Version A.

5-A. Classifying Tables | Version A

Classify each table: Table 1 y-values: 5, 12, 19, 26 Table 2 y-values: 2, 6, 18, 54 Table 3 y-values: 3, 4, 7, 12 Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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5-B. Classifying Tables | Version B

Classify each table: Table 1 y-values: 8, 15, 22, 29 Table 2 y-values: 4, 10, 25, 62.5 Table 3 y-values: 1, 3, 7, 13 Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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6-A. Writing $y = ab^x$ | Version A

Write an exponential equation for x: 0, 1, 2, 3 and y: 7, 21, 63, 189. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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6-B. Writing $y = ab^x$ | Version B

Write an exponential equation for x: 0, 1, 2, 3 and y: 12, 9.6, 7.68, 6.144. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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7-A. Growth and Decay | Version A

For $y = 100(1.15)^x$: a. Growth or decay? b. Percent rate? Answer: _____	Work / thinking _____ _____ _____ _____ _____
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7-B. Growth and Decay | Version B

For $y = 50(0.82)^x$: a. Growth or decay? b. Percent rate? Answer: _____	Work / thinking _____ _____ _____ _____ _____
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8-A. Depreciation | Version A

Equipment costs \$8,500 and depreciates 18% per year. Write a model and find the value after 5 years. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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8-B. Depreciation | Version B

Equipment costs \$12,000 and depreciates 12% per year. Write a model and find the value after 4 years. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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9-A. Appreciation | Version A

A trading card costs \$450 and appreciates 22% per year. Write a model and find the value after 8 years. Answer: _____	Work / thinking
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9-B. Appreciation | Version B

Furniture costs \$3,200 and appreciates 6% per year. Write a model and find the value after 10 years. Answer: _____	Work / thinking
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10-A. Linear Prediction | Version A

A linear model for study hours and score is $y = 8.8x + 54$. Predict the score for 6.5 hours. Answer: _____	Work / thinking
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10-B. Linear Prediction | Version B

A linear model for miles and gallons is $y = 0.05x$. Predict gas used for 350 miles. Answer: _____	Work / thinking
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11-A. Exponential Prediction | Version A

A bacteria model is $y = 8000(0.7)^x$. Predict the bacteria count after 6 hours. Answer: _____	Work / thinking
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11-B. Exponential Prediction | Version B

A medicine model is $y = 250(0.8)^x$. Predict the concentration after 5 hours. Answer: _____	Work / thinking
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Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the setup, notation, and calculation.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

B. $y = 6(4)^x$	A. 12% growth
A. 128	C. exponential
C. 225	D. 24% decay
B. $y = 20(0.5)^x$	B. quadratic

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

Table 1 linear; Table 2 exponential; Table 3 quadratic	$V(t) = 3200(1.06)^t$; $V(10) \approx \$5,730.71$
$V(t) = 12000(0.88)^t$; $V(4) \approx \$7,196.34$	17.5 gallons
$y = 7(3)^x$	Table 1 linear; Table 2 exponential; Table 3 quadratic
$V(t) = 450(1.22)^t$; $V(8) \approx \$2,208.47$	111.2 points
$V(t) = 8500(0.82)^t$; $V(5) \approx \$3,151.29$	≈ 941 bacteria
a. decay; b. 18%	81.92 mg/L
a. growth; b. 15%	$y = 12(0.8)^x$

Final Check Before the Assessment

- ☐ I can classify tables as linear, quadratic, or exponential.
- ☐ I can identify a and b in $y = ab^x$.
- ☐ I can tell whether an exponential model shows growth or decay.
- ☐ I can write and use exponential growth/decay models.
- ☐ I can use linear and exponential models to make predictions.